Control System Lab
Mid Exam

تم حل هذا الملف من قبل الطالب : ليث عمرو

هذه الحلول هي جهد بشري قابل للخطأ، فإن أحسنت فمن الله
وإن أخطأت فمنّي ومن الشيطان.

هذا العمل مقدّم ثوابه لروح والدي (قاسم مفيد عمرو)
فمن وجد فيه فائدة فليدعُ له بالرحمة والمغفرة.

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Control System Lab
Mid Exam

Q1) Using the following system.

A) Determine the state-space matrices A, B, C, and D for a system given by:

$$y''(t) + 0.6y'(t) + 2y(t) = 3r(t)$$

A [ ]

B [ ]

C [ ]

D [ ]

Solution:

A [0 1 ; -2 -0.6]

B [0 ; 3]

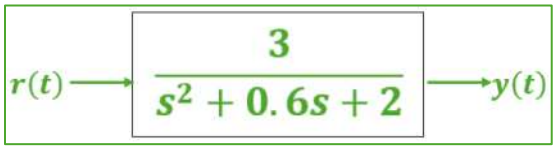
C [1 0]

D [0]

B) Find the transfer function of above equation, and draw the simulation diagram.

Solution:

$$\frac{Y(s)}{R(s)} = \frac{3}{s^2 + 0.6s + 2}$$

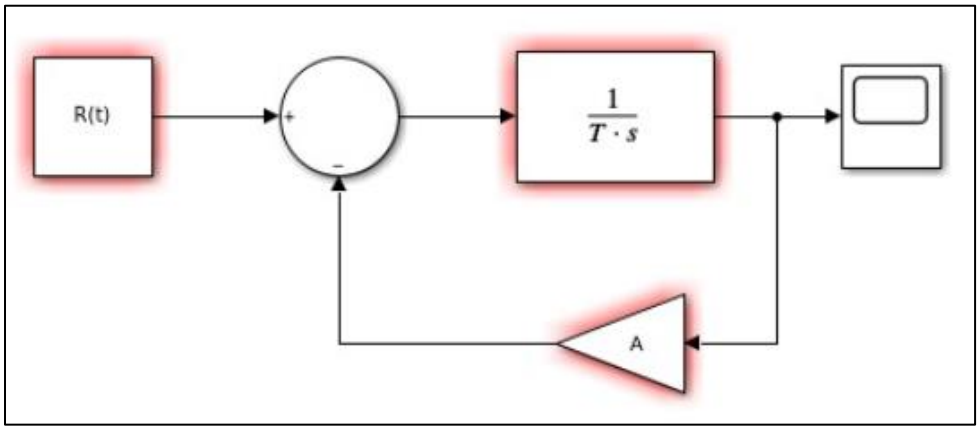


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Control System Lab
Mid Exam

Q2)

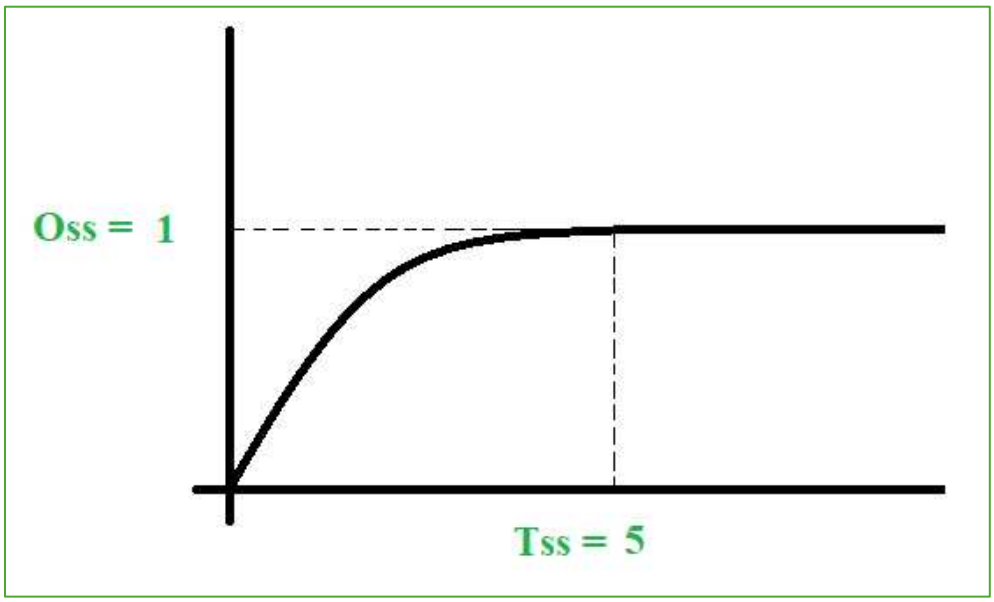
A) Plot the output response for each for the following conditions.



1. $R(t) = 1, T = 1, A = 1$
2. $R(t) = 1,\ T = 1,\ A = 0$
3. $R(t) = 1,\ T = 3,\ A = 1$
4. $R(t) = 1,\ T = 1,\ A = 4$
5. $R(t) = 2,\ T = 1,\ A = 1$

Solution:

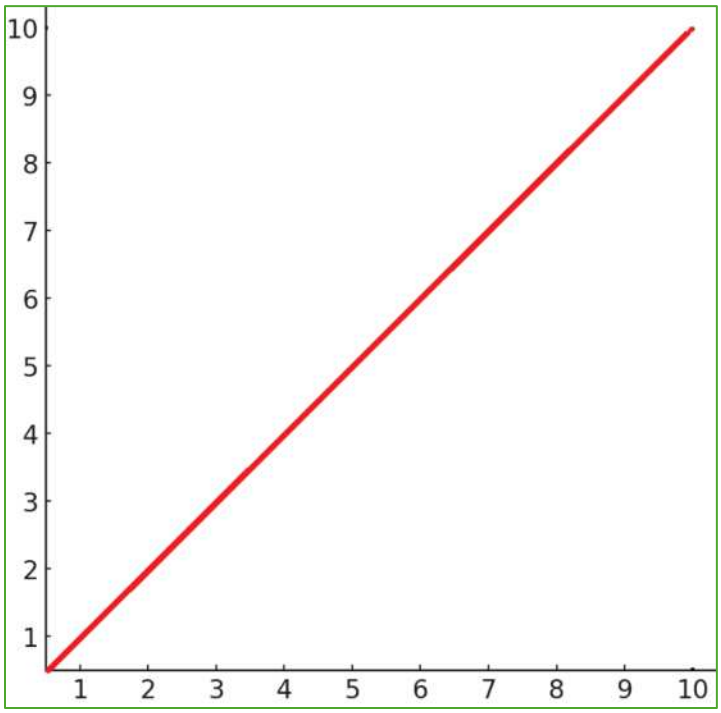
1.



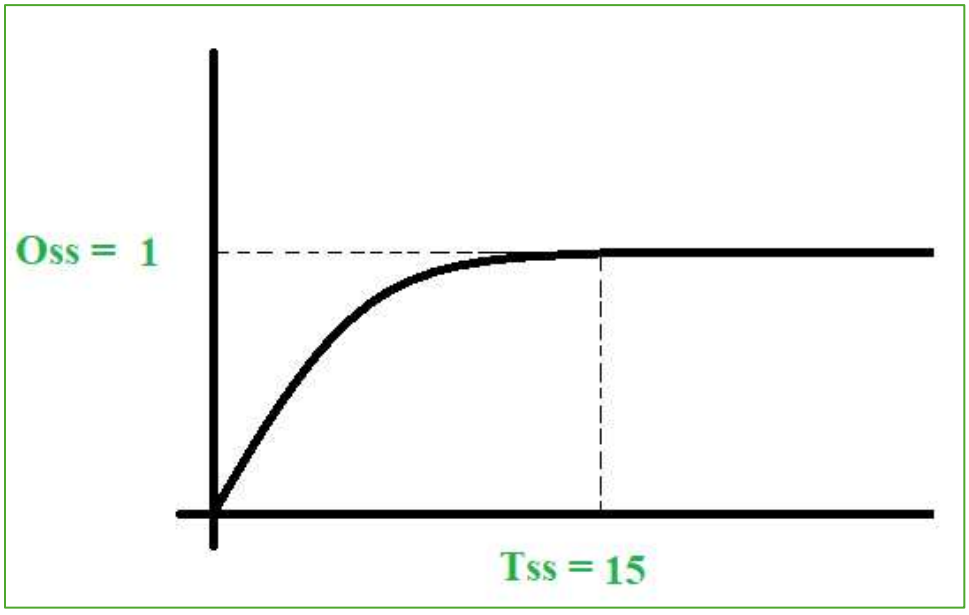
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Control System Lab
Mid Exam

2.



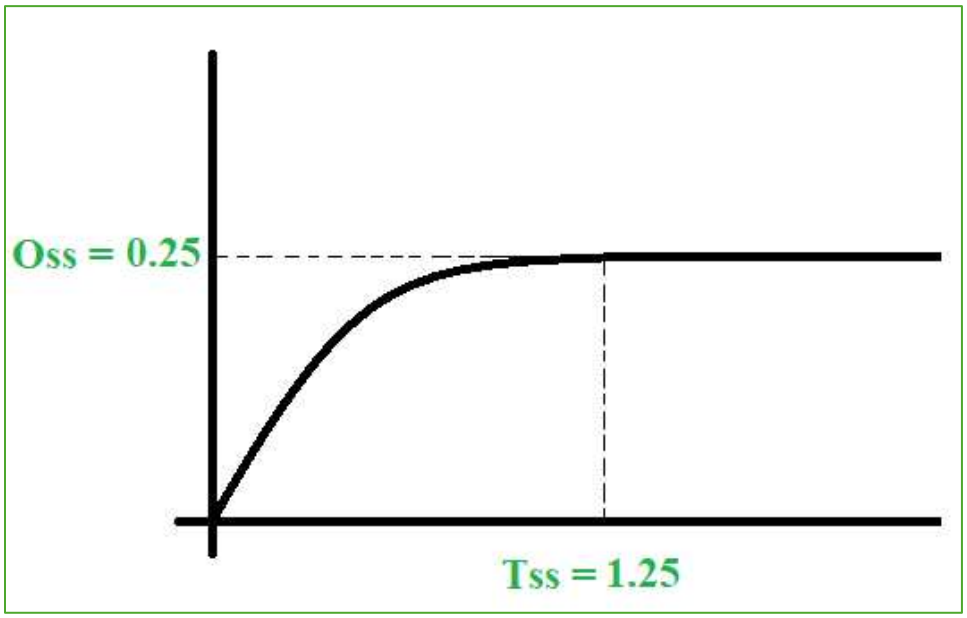
3.



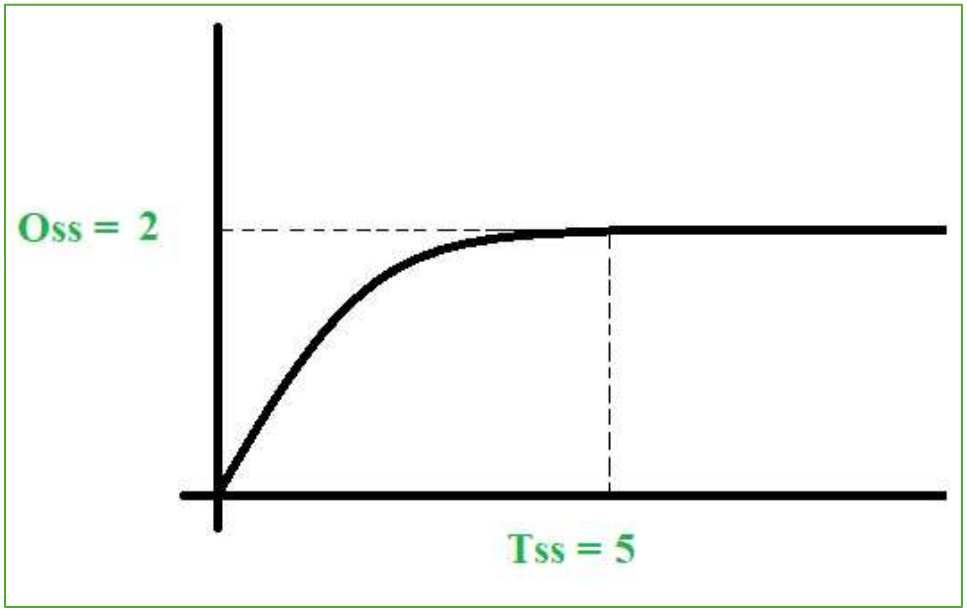
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Control System Lab
Mid Exam

4.



5.



Control System Lab
Mid Exam

B) Explain the effect of increased input in the first order system.

Solution:

The final output in steady state increases, same response speed.

C) Explain the effect of increased T in the first order system.

Solution:

Lower system response, long time to reach steady state.

D) Explain the effect of negative fixed feedback on time constant and gain.

Solution:

Time constant → Decreased
Gain → Decreased

E) Explain the effect of negative flexible feedback on time constant and gain.

Solution:

Time constant → Increased
Gain → No change

F) Explain the effect of positive fixed feedback on time constant and gain.

Solution:

Time constant → Increased
Gain → Increased

G) Explain the effect of positive flexible feedback on time constant and gain.

Solution:

Time constant → Decreased
Gain → No change

Control System Lab

Mid Exam

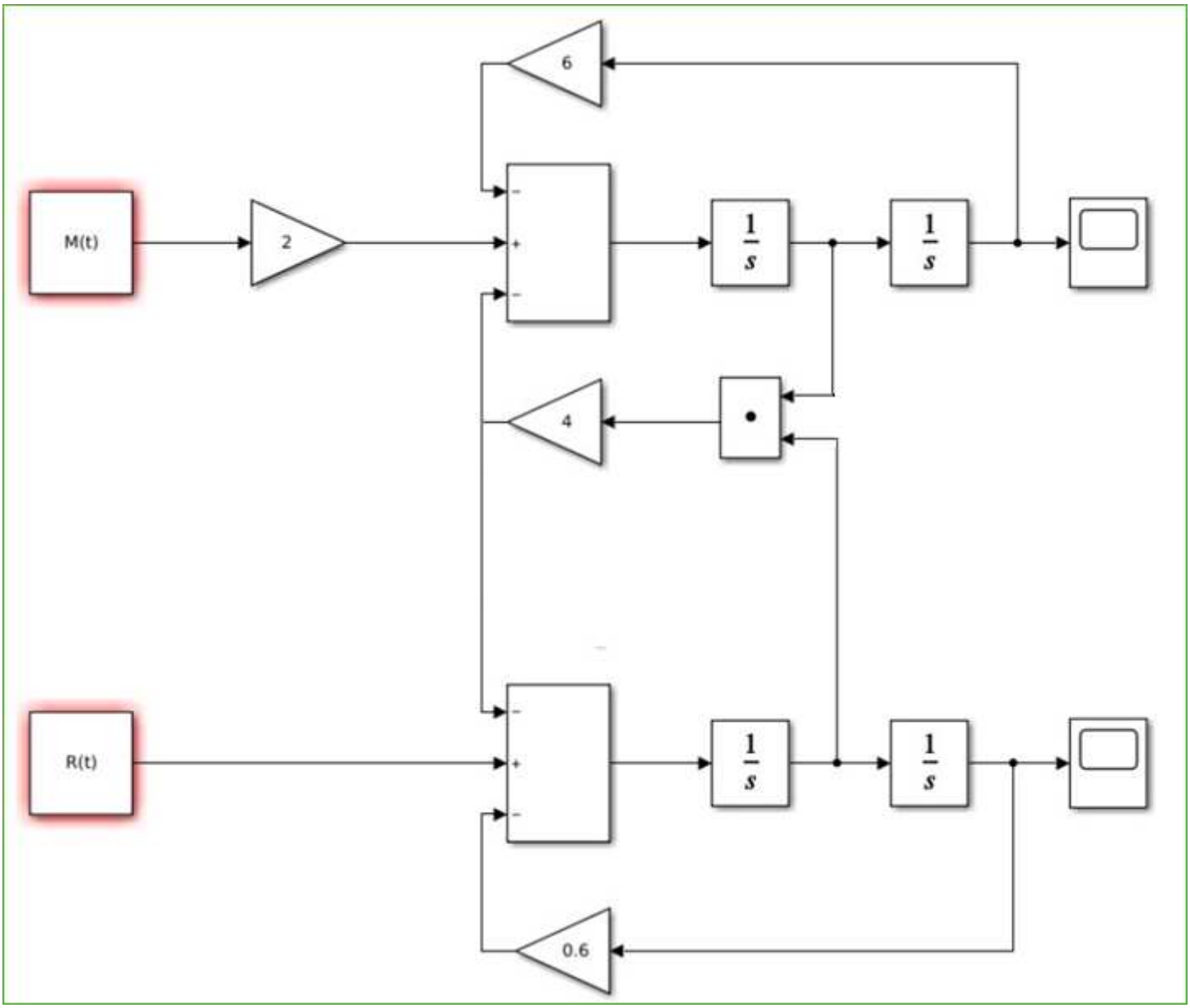
Q3) Represent the following systems by integration in Simulink:

$$M(t) = 0.5Y''(t) + 2Y'(t)Z'(t) + 3Y(t)$$

$$R(t) = Z''(t) + 4Z'(t)Y'(t) + 0.6Z(t)$$

Solution:

$$Y''(t) = 2M(t) - 4Y'(t)Z'(t) - 6Y(t)$$
$$Z''(t) = R(t) - 4Z'(t)Y'(t) - 0.6Z(t)$$



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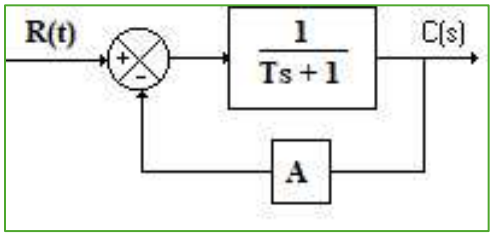
Control System Lab
Mid Exam

Q4) Practical (عملي):

A) Simulate the first order of closed loop system, with negative feedback $H(s) = A$.

$$R(t) = 2 \,/\, T = 0.6, 1, 3 \,/\, A = 1$$

Solution:



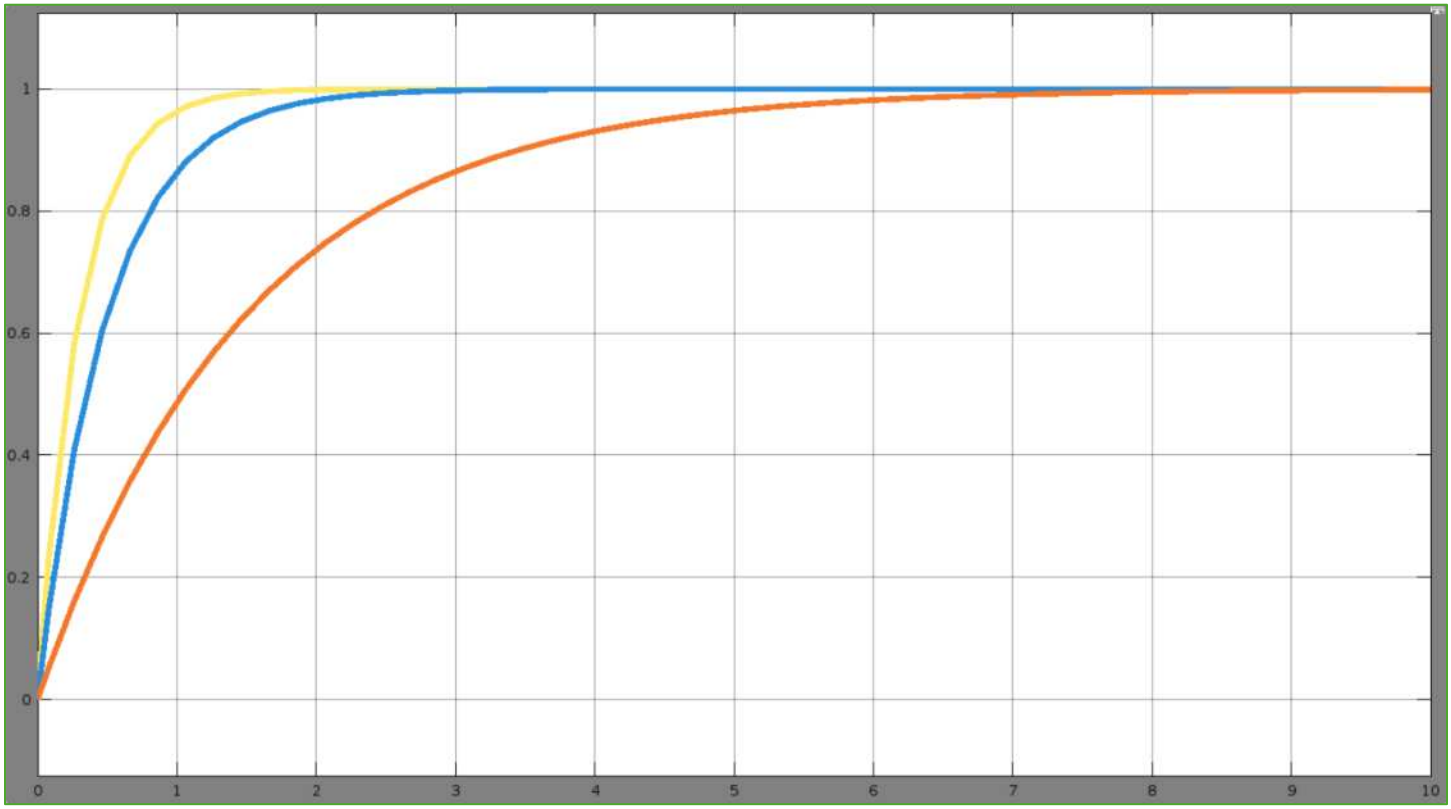
B) Record the output in table.

Solution:

T	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0	5.5	6.0
0.6	0.8	0.95	0.99	1	1	1	1	1	1	1	1	1
1.0	0.6	0.87	0.94	0.97	0.99	1	1	1	1	1	1	1
3.0	0.27	0.48	0.63	0.73	0.82	0.88	0.90	0.92	0.93	0.95	0.97	0.98

C) Plot the Output.

Solution:



D) Study the result.

Solution:

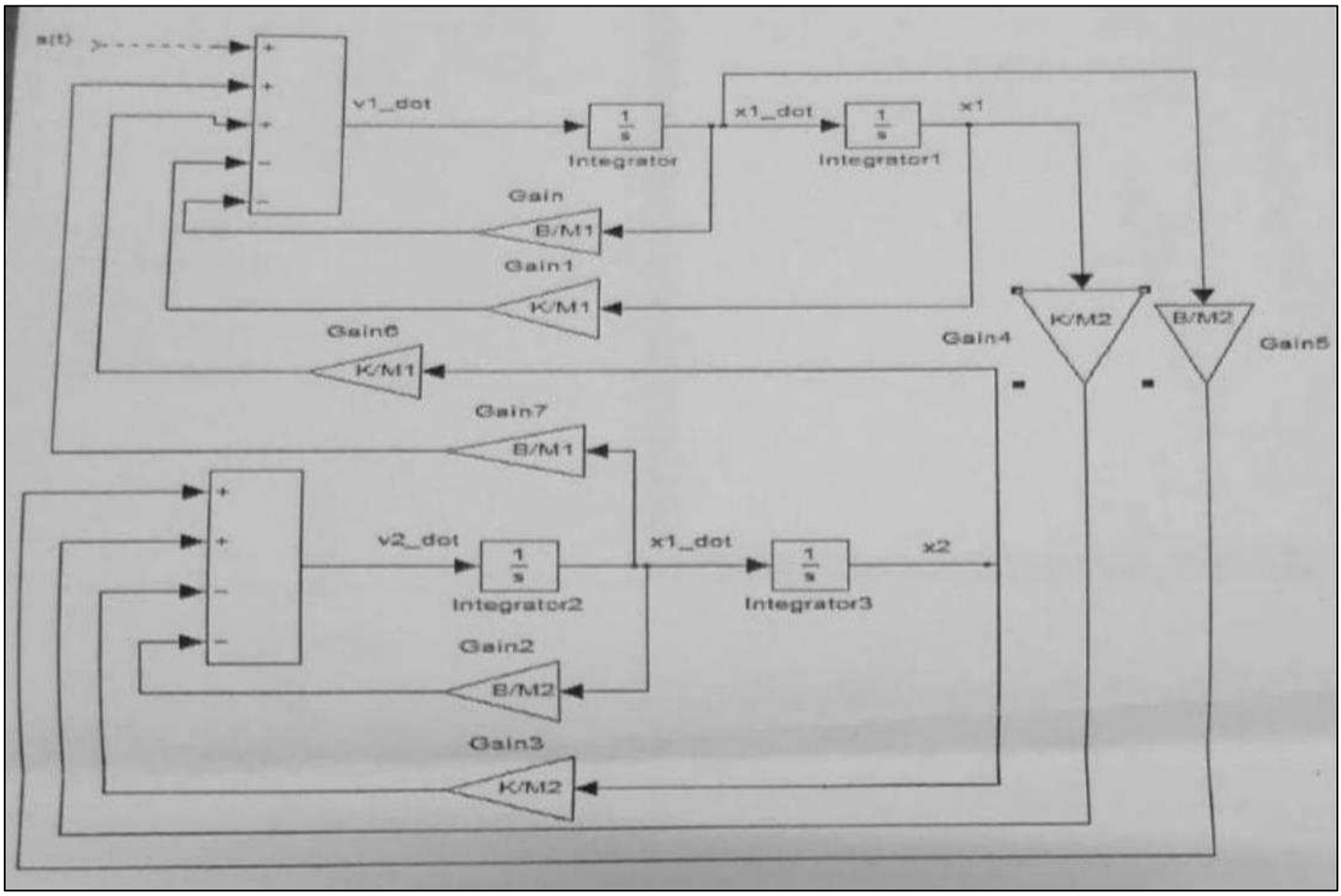
When T increases the settling time of system will increase (with same feedback).

When T increases the response speed of system will decrease (with same feedback).

Control System Lab

Mid Exam

Q5) Write the differential equations V1_dot & V2_dot for the following system:



Solution:

$$V1_dot = a(t) + \left(x1_dot * \frac{B}{M1}\right) + \left(x2 * \frac{K}{M1}\right) - \left(x1 * \frac{K}{M1}\right) - \left(x1_dot * \frac{B}{M1}\right)$$

$$V2_dot = \left(x1_dot * \frac{B}{M2}\right) + \left(x1 * \frac{K}{M2}\right) - \left(x2 * \frac{K}{M2}\right) - \left(x1_dot * \frac{B}{M2}\right)$$

تم بحمد الله.

اللهمَّ ارحم عبدَك (قاسم مفيد عمرو) واجعله من ورثة جنّة النعيم.

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